

Article

Baby Cry Classification Using Structure-Tuned Artificial Neural Networks with Data Augmentation and MFCC Features

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Abstract: Babies express their needs, such as hunger, discomfort, or sleeplessness, by crying. However, understanding these cries correctly can be challenging for parents. This can delay the baby's needs, increase parents' stress levels, and negatively affect the baby's development. In this paper, an integrated system for the classification of baby sounds is proposed. The proposed method includes data augmentation, feature extraction, hyperparameter tuning, and model training steps. In the first step, various data augmentation techniques were applied to increase the training data's diversity and strengthen the model's generalization capacity. The MFCC (Mel-Frequency Cepstral Coefficients) method was used in the second step to extract meaningful and distinctive features from the sound data. MFCC represents sound signals based on the frequencies the human ear perceives and provides a strong basis for classification. The obtained features were classified with an artificial neural network (ANN) model with optimized hyperparameters. The hyperparameter optimization of the model was performed using the grid search algorithm, and the most appropriate parameters were determined. The training, validation, and test data sets were separated at 75%, 10%, and 15% ratios, respectively. The model's performance was tested on mixed sounds. The test results were analyzed, and the proposed method showed the highest performance, with a 90% accuracy rate. In the comparison study with an artificial neural network (ANN) on the Donate a Cry data set, the F1 score was reported as 46.99% and the test accuracy as 85.93%. In this paper, additional techniques such as data augmentation, hyperparameter tuning, and MFCC feature extraction allowed the model accuracy to reach 90%. The proposed method offers an effective solution for classifying baby sounds and brings a new approach to this field.

Keywords: baby cry classification; mel-frequency cepstral coefficients; data augmentation; grid search optimization; artificial neural network

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1. Introduction

Babies cry as an innate reflex to express their needs, such as hunger, discomfort, and sleeplessness. Understanding these cries accurately and quickly is critical to babies' healthy development and happiness [1]. However, inexperienced parents, new caregivers, and some healthcare professionals may have difficulty understanding why babies cry. This can lead to prolonged crying, increased stress levels, and negatively impacted cognitive development [2].

Studies on the automatic classification of baby crying sounds have significantly contributed to this field. Analysis and classification of baby crying sounds using sound processing methods and machine learning algorithms constitute the basis of research in this

field. This literature review examines important studies on the classification and detection of baby crying sounds. It discusses, in detail, the methodologies, datasets, feature extraction methods, and classification algorithms used in these studies.

Various studies on the classification of baby cries have developed different methods using sound processing methods and machine learning algorithms. Sharma [3] classified the conditions such as hunger, need to burp, abdominal pain, discomfort, fatigue, loneliness, feeling cold or hot, and fear on the Donate a Cry dataset and achieved 81.27% accuracy with k-means, hierarchical clustering, and Gaussian mixture models using features such as mean frequency, median frequency, spectral entropy, and skewness.

Maghfira [4] achieved 94.97% success in classifying hunger, discomfort, need to pass gas, and abdominal pain using the Dunstan Baby dataset with CNN-RNN. Similarly, Franti [5] achieved 89% accuracy with CNN on the same dataset. Liu [6] achieved 76.4% success with the nearest neighbor algorithm and artificial neural networks using LPC, LPCC, MFCC, and BFCC features on sounds recorded in the NICU environment. Turan [7] achieved 86.1% accuracy with capsule networks' help on the CRIED dataset.

Osmani [8] performed classification using SVM, Bagging Decision Tree, and Boosted Trees algorithms using parameters such as spectrum, tone, zero crossing rate, root mean square, energy, and statistical features in the Dunstan Baby dataset. Chang [9] evaluated spectrogram-based features using CNN to classify hunger, pain, and sleep categories on the dataset collected from National Taiwan University Hospital with an accuracy rate of 78.5%. Bano [10] achieved 86% accuracy with KNN using features such as MFCC and short-term energy on its own recorded data. Orlandi [11] used features such as median, mean, standard deviation, and minimum and maximum values of F0 and F123 parameters to distinguish full-term and preterm babies and achieved 87% success with logistic regression, multilayer perceptron, SVM, and RF algorithms. Bhagatpatil [12] achieved 91.58% accuracy with K-means clustering and KNN algorithms using MFCC and LFCC features on his own recorded data. Rosales-Pérez [13] achieved 97.96% accuracy by classifying hunger and pain states on the Baby Chillanto dataset using MFCC and LPC features using a fuzzy model algorithm. Yamamoto [14] achieved 62.1% accuracy with the nearest neighbor algorithm using FFT features on his own recorded discomfort, hunger, and sleep categories. Finally, Bashiri and colleagues [15] achieved 99.9% accuracy using MFCC and artificial neural network (ANN) algorithms with the BabyChilanto dataset.

Various studies on baby crying detection have developed different approaches to overcome the challenges in this area using sound processing methods and machine learning algorithms. For example, Chang [16] performed baby crying detection using a self-recorded dataset. The data included background sounds such as television and speech, and 99.83% accuracy was achieved using the CNN algorithm with spectrogram-based features. Manikanta [17] extracted MFCC features by recording baby crying data accompanied by the air conditioner and fan sounds in a home environment and achieved an 86% accuracy rate using 1D-CNN, FFNN, and SVM algorithms. Dewi [18] achieved 90% accuracy with the KNN algorithm using LFCC features on a dataset containing baby crying and non-crying sounds that she recorded. Gu [19] used LPC features on a dataset containing background sounds such as laughter and dog barking that she recorded and achieved a 97.1% success rate with the Dynamic time warping algorithm. Ferretti [20] worked on two different datasets: the first is real data recorded in a hospital's Neonatal Intensive Care Unit (NICU); the second is a dataset containing synthetic sounds such as speech and "beep" sounds. The CNN algorithm with Log-Mel coefficients features was used, and an accuracy rate of 86.58% was achieved in the real dataset and 92.92% in the synthetic dataset. Feier [21] worked on the TUT Rare Sound Events 2017 dataset. This dataset includes sounds such as glass breaking, gunshots, and baby cries. Using log-amplitude mel-spectrogram features, CRNN

achieved an 85% success rate for baby cry detection and 87% for all targets. Torres [22] worked on a dataset containing background sounds, such as adult crying and vacuum cleaners, collected from online sources. Features such as MFCC, voiced–unvoiced counter, and harmonic ratio accumulation with consecutive F0 were extracted from the audio data. A 92% AUC rate was achieved with the Support Vector Data Descriptor (SVDD) and CNN algorithms. Lavner [23] worked on a dataset containing background sounds such as talking and door opening in a home environment. Using the CNN algorithm with features such as MFCC, pitch, and formant, 95% accuracy was achieved.

This study aims to develop a system that can automatically classify the needs or distresses underlying the crying sounds of babies by analyzing them. The system will extract meaningful data using sound processing techniques and machine learning algorithms and classify this data, allowing parents and caregivers to respond to babies' needs accurately and quickly. Sound processing methods such as Mel-Frequency Cepstral Coefficients (MFCCs) will be used in the research. The development of this system will support healthy development by reducing babies' stress levels. In addition, it will also contribute to the establishment of a healthier and happier relationship by reducing the stress of parents and caregivers. In addition, this study will contribute to the academic literature on sound processing and machine learning and will guide parents as a practical training and support tool.

The main contributions of the article are as follows:

- The study provides a more effective solution than existing approaches in this field by developing an ANN-based integrated model for classifying baby sounds.
- The artificial neural network model used in the study was improved by hyperparameter optimization using the grid search optimization algorithm, and its accuracy rate was significantly increased.
- To the best of our knowledge, the proposed model, which has the best average accuracy rate of 90%, outperforms the existing work [24] that reported an 85.93% accuracy on Donate a Cry Dataset test data.

The rest of this paper is organized as follows. Materials and Methods are presented in Section 2. Section 3 describes the experimental studies, including the experimental setup and results. Finally, Section 4 summarizes the conclusion and future work.

2. Materials and Methods

2.1. Experimental Setup

Hardware and software setups are explained in detail in the following subsections.

2.1.1. Hardware Setup

Deep learning training was performed on a computer with the following specifications: CPU (Intel Xeon 3.30 GHz), 128 GB RAM, and GPU is a NVIDIA (Santa Clara, CA, USA) with CUDA Toolkit version 11.2 (cu112).

2.1.2. Software Installation

Python (version 3.8.20) programming was used to create artificial neural network models to classify baby sounds. Python's flexible structure and extensive library support facilitated artificial intelligence and data processing. TensorFlow (version 2.13.0) and Keras (version 2.13.1) libraries were used to classify baby sounds, create models, and train processes. While TensorFlow (Google LLC, Mountain View, CA, USA) accelerated the model's training process with its high-performance computing infrastructure, Keras (TensorFlow, Mountain View, CA, USA) facilitated the model's design, hyperparameter optimization, and evaluation processes with its user-friendly interface. In addition, the

Librosa library (version 0.9.2) was used to process and analyze sound data. With the help of Librosa (RRZ Labs, San Francisco, CA, USA), loading and normalizing sound files and extracting important sound features such as Mel spectrogram and MFCC were provided.

Versions of the libraries and programming language used are shown in Table 1.

Table 1. Software versions used in the study.

Software	Version
Python	3.8.20
Tensorflow	2.13.0
Keras	2.13.1
Numpy	1.24.3
Matplotlib	3.7.5
Seaborn	0.13.2
Scipy	1.10.1
Scikit-Learn	1.3.2
Pandas	2.0.3
Librosa	0.9.2

2.2. Dataset

The studies were conducted on Donate a Cry, a publicly available dataset.

2.2.1. Donate a Cry Dataset

The Donate a Cry dataset was developed through recordings of infants' crying sounds and relevant details collected by parents using a mobile application compatible with both iOS and Android devices. Users who uploaded audio recordings followed specific instructions to label the cries and provided the presumed reasons for crying. Initially published by Gaber Veres on GitHub in 2015 [25], the dataset underwent updates involving pre-audio processing for each recording. The dataset comprises audio files of infant cries from babies aged between 1 and 22 months. The revised version of the dataset was utilized in this research. Table 2 illustrates the distribution of data across different classes. For a fair comparison, the dataset was preprocessed to have the same number of data and the same type of splitting as the work in the literature. Data splitting and data augmentation are presented in detail in the following subsections.

Table 2. Classes and record counts in the Donate a Cry dataset.

Class	Record
Belly pain	16
Burping	8
Discomfort	27
Hungry	382
Tired	24
Total	457

2.2.2. Data Splitting

The splitting phase is critical to understanding the underlying structure of the dataset. Before proceeding to the preprocessing phase, the data were divided into three subsets: training set 75%, validation set 10%, and test set 15%. The determination of these subsets was conducted to provide an unbiased assessment of the model's performance and to make a fair comparison with the studies in the literature.

The audio files were augmented before classification. During the Augment process, the newly created audio files were recorded by adding the aug abbreviation to their file

names. When separating the training and test audio, different audio files in the class were separated according to their file names.

2.3. Data Preprocessing

Data augmentation and feature extraction were applied to the dataset for preprocessing and given in detail in the following subsections.

2.3.1. Data Augmentation (DA)

Audio data augmentation methods use different techniques to increase the model's generalization ability and add variety to the dataset. The following data augmentation techniques were implemented using the Librosa tool [26] considered in the following:

1. **Time Shift:** Shifting the audio signal in time by a certain amount (positively or negatively) makes the model insensitive to the location of the sound. This technique is used when information dependent on the timing of the sound should not be important to the model and makes the model robust to sound samples at different timings.
2. **Time Stretch:** Changing the length of the sound without changing its speed, that is, speeding it up or slowing it down, allows the model to be robust to different speeds of sound. This helps the model be flexible to different tempos of sound and allows it to make accurate predictions despite changes in speed.
3. **Pitch Shift:** Changing the pitch of the sound, i.e., frequency shifting, makes the model robust to different tones. Pitch shifting allows models that are not particularly sensitive to changes in pitch to be more generalizable to a wide range of tones.
4. **Add White Noise:** Adding a low level of random white noise to the audio signal allows the model to perform better in noisy environments. This technique helps the model better adapt to real-world environments, except when the audio is clean.
5. **Slice:** Generating shorter audio samples by randomly taking a section from a long audio file allows the model to focus on specific audio segments. This allows the model to learn the information in long audio files more effectively and focus on important details in specific short segments.

The parameters of the methods used are shown in Table 3.

Table 3. Augmentation parameters for Donate a Cry.

Method	Technique	Parameter	Value	Description
Time Shift	Shifting	Samples	−500 to 500	Random scrolling
Time Stretch	Speed Change	Ratio	0.8 to 1.2	Random speed change
Pitch Shift	Pitch Change	Semitones	+4	Shift up
Add Noise	Gaussian Noise	Mean	0	White noise
		Variance	0.000025	Low-level noise
		Power	0.005	0.5% of signal amplitude
Slice	Segmentation	Duration	1 s	Cut a random segment

2.3.2. Data Resizing

Making audio files of different lengths equal ensures that the model receives each audio sample of the same size, creating a more consistent training process. When audio files have different lengths, the model often needs to adjust the audio files to a certain length in order to correctly process each sample. This can be done by trimming the end of the audio file or padding the missing portion.

This method was used in the study, especially when the lengths of the audio files were different. If the length of the audio file is shorter than the specified size, the audio file is

completed to the specified length by adding zeros at the end (padding). If the length of the audio file is longer than the specified size, the excess part of the audio file is cut off (trimming). This helps the model to provide a consistent training process by taking each audio sample as the same size.

- **Trimming** It is the shortening of an audio file to a certain length. In long audio files, parts that exceed the specified maximum length are cut off. This is usually done on the assumption that the important part of the audio is at the beginning.
- **Padding.** The process of adding zero or a constant value to short audio files until they reach a certain length. This is especially useful when the dataset contains very short audio files. Padding can be added to the end of the audio or spread out over a certain period of time.

The minimum, maximum, and average durations of the records in the dataset based on classes are shown in Table 4.

Table 4. Duration for Donate a Cry.

Class	Min. Duration (s)	Max. Duration (s)	Avg. Duration (s)
Belly pain	5.70	8.19	6.93
Burping	6.40	7.94	6.94
Discomfort	5.87	8.28	6.95
Hungry	6.52	7.06	6.91
Tired	5.77	8.54	6.87

Artificial neural networks (ANNs) usually work with fixed-length input vectors. However, in data with variable duration, such as audio data, the length of each record may be different. This can be a problem for ANN to get input vectors because the network needs to work with a fixed length vector that it expects. Therefore, padding and trimming methods are applied to make audio files of equal length.

2.3.3. Feature Extraction

A comprehensive literature review was conducted to improve the feature extraction process. Important studies by Chunyan Ji [27] were examined in detail, and feature selection was made in light of these studies. Among the selected features from the cepstral domain are Mel-Frequency Cepstral Coefficients (MFCCs). Zero crossing ratio (ZCR) and root mean square (RMS) features from the time domain were also considered, but only the MFCC feature was used in this study.

Example MFCC images of each class are shown in Figure 1.

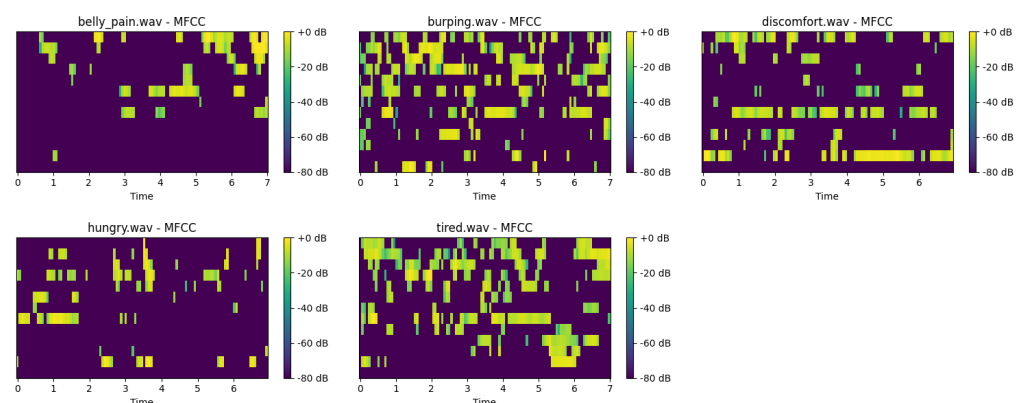


Figure 1. Mel spectrogram example for each class.

The step-by-step process for computing MFCC is illustrated in the block diagram in Figure 2. The block diagram presented in Figure 2 is given to explain the general operation of the MFCC calculation process.

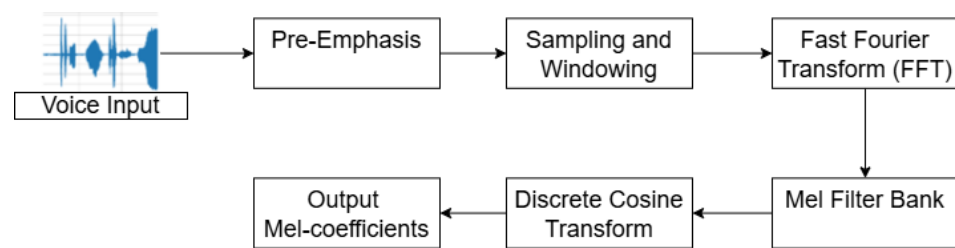


Figure 2. Block diagram of the MFCC computation.

1. **Pre-Emphasis:** The initial speech signal $x(n)$ is passed through a high-pass filter to enhance higher frequency components [28]:

$$y(n) = x(n) - a \cdot x(n-1) \quad (1)$$

Here, $y(n)$ represents the filtered signal, and the coefficient a typically ranges between 0.9 and 1.0. This step emphasizes the higher frequencies, increasing their energy for improved feature extraction [29].

2. **Frame Blocking:** The speech signal is typically divided into overlapping frames, each lasting between 20 and 30 ms. Frames have a certain amount of overlap, and the frame length is usually greater than this overlap. Commonly used values for frame length and overlap are 256 and 100, respectively [29].
3. **Windowing:** Windowing is applied to reduce signal discontinuities at frame edges. A smooth window function, such as the Hamming window, is used, which tapers to zero at its extremes. This approach maintains signal continuity between frames. The Hamming window is defined as Formula (2):

$$Y(n) = X(n) \cdot W(n) \quad (2)$$

Here, $W(n)$ is the window function [29].

4. **Fast Fourier Transform (FFT):** FFT converts the time domain signal into the frequency domain, facilitating feature extraction while retaining all essential information. The magnitude spectrum is smoothed using triangular band-pass filters to simplify speech processing [30].
5. **Discrete Cosine Transform (DCT):** DCT is applied to the outputs of the band-pass filters to compute the Mel-Frequency Cepstral Coefficients (MFCCs). The transformation from the frequency domain back to the time domain is given by Formula (3):

$$C(n) = \sum E_k \cdot \cos\left(n \cdot (k-0.5) \cdot \frac{\pi}{40}\right) \quad (3)$$

Here, n ranges from 0 to N , N represents the number of triangular band-pass filters, and L is the number of Mel-scale Cepstral Coefficients.

The k value is the index of the band-pass filters and usually represents the number of frequency components. This value indicates the order of the band-pass filters used. These coefficients are widely used in speech recognition tasks [30].

In this study, certain parameters are used to extract Mel-Frequency Cepstral Coefficient (MFCC) features from audio signals. The sampling frequency (sample rate) is selected as 16,000 Hz. This selection is a common standard, especially in speech and general audio processing applications and provides a sufficient resolution representation of the sound.

The number of MFCC coefficients is determined as 13. Generally, the first few MFCC coefficients represent the basic spectral features of the sound, while higher coefficients capture fine details. Using 13 coefficients is a balanced choice in terms of both preventing information loss and keeping the processing load at a reasonable level.

In MFCC calculation, instead of coefficients that vary over time, it is preferred to take average values. In this way, the effect of short-term fluctuations in the audio signal is reduced, and a more stable feature vector is obtained. In addition, by using mel-frequency filter banks and discrete cosine transform (DCT), the frequency distribution perceived by the human ear is better modeled, and the distinctive features of the sound are extracted more effectively. The selection of these parameters is made in order to minimize the information loss while characterizing the audio signals and increasing the classification performance. Before feeding the MFCCs to the network, the features are vectorized.

Popular Python libraries such as Librosa, Scipy, Numpy, and Pandas played a pivotal role in implementing the feature extraction and normalization steps, enabling an efficient and streamlined process.

2.4. Artificial Neural Network Methods

Artificial neural networks (ANNs) are computational models derived from the architecture and operation of the neural networks in the human brain. These models simplify biological neural processes to create interconnected systems capable of various tasks [31]. ANNs, often referred to as neural networks or direct neural networks in academic and engineering contexts, consist of numerous nodes (neurons) that work together to process information [32].

Each node in the network performs a specific output function, commonly known as the activation function. The connections between nodes are associated with weights, which determine the signal strength transmitted through these links. These weights are akin to the memory of the neural network and play a crucial role in its learning process [33]. The overall output of the network depends on the structure of its connections, the assigned weight values, and the chosen activation functions. ANNs can approximate complex algorithms or functions, often representing logical or strategic operations [34].

The architecture of artificial neural networks (ANNs) is typically organized into three main layers: the input layer, hidden layer(s), and output layer. The input layer serves as the interface between the network and the external environment, receiving signals and data for processing [35]. The hidden layer(s), situated between the input and output layers, handles the intermediate processing of the received data. These layers are not directly observable from the outside but play a crucial role in extracting features and learning patterns [36]. Finally, the output layer generates the network's results, delivering the processed information to the user or an external system. By modifying the connectivity between nodes, adjusting the weights, and selecting appropriate activation functions, ANNs can be tailored to perform a wide range of tasks, from basic computations to tackling complex, abstract problems.

This study focuses on developing a model based on artificial neural networks (ANNs). The first stage of the model starts with a basic ANN architecture. The building blocks of this basic model are as follows:

- Input Layer: The layer where the data will be introduced to the model.
- Five Hidden Layers: Layers that allow the model to learn more complex relationships. These layers contain the following components:
 - Linear Transformation: Processing the data in a linear manner.
 - Normalization: Normalization is conducted to process the data in a standard manner.

- ReLU Activation: The ReLU (rectified linear unit) activation function is a function that sets negative inputs equal to zero and leaves positive inputs as they are:

$$f(x) = \max(0, x) \quad (4)$$

It is frequently used in deep learning models thanks to its simplicity and linear piecewise structure.

- Dropout Layer: Some neurons are randomly turned off to prevent the model from overfitting.
- Output Layer: This is the last layer where the model produces the results, and here, the LogSoftmax activation function is used. This function normalizes the probabilities of each class by taking the logarithm of the outputs.

Formula:

$$\text{LogSoftmax}(z)_i = \log\left(\frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}}\right) \quad (5)$$

Here:

- Vector z is the output, which is the classification score of the model;
- i represents each element of these scores;
- K represents the total number of classes.

2.5. Proposed Method

The primary objective of this study is to develop an effective integrated classification system to distinguish different types of baby crying sounds. The proposed method consists of four main phases: data augmentation, feature extraction, hyperparameter tuning, and model training. The overall architecture of the proposed method is illustrated in Figure 3.

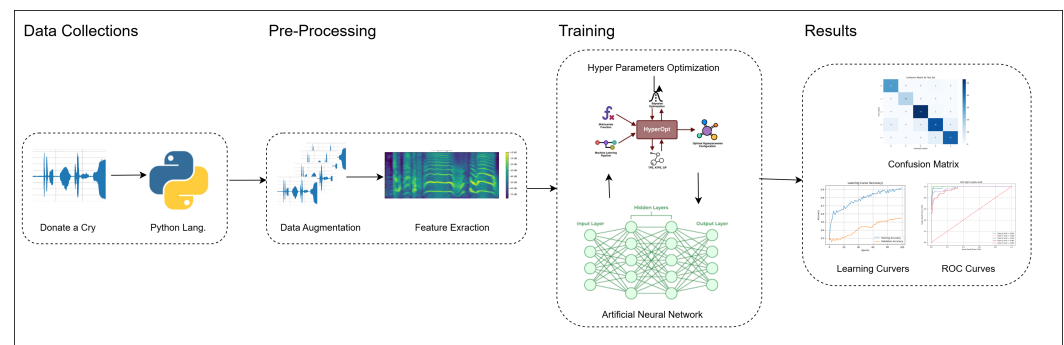


Figure 3. The architecture of the proposed method.

Phase 1: Data Augmentation. In order to improve the generalization ability of the model and prevent overfitting, data augmentation techniques are applied to the raw audio signals. Augmentation is particularly important in deep learning applications where the amount of available training data is limited [37]. In this phase, data augmentation is applied to audio data using time shifting, time stretching, pitch shifting, adding white noise, and slice methods. Data augmentation was performed for the entire set. While splitting, care was taken to ensure that the real and augmented data were not present in both the train and test sets. More details about the data augmentation techniques are provided in Section 2.3.1.

Phase 2: Feature Extraction Mel-Frequency Cepstral Coefficients (MFCCs) are widely used in speech and sound processing because they effectively capture the spectral characteristics of an audio signal in a way that is similar to human auditory perception. In this phase, MFCCs are extracted from the augmented audio samples. The extracted MFCC

features serve as input to the classification model. Further details on feature extraction can be found in Section 2.3.3.

Phase 3: Hyperparameter Tuning To optimize the classification performance of the artificial neural network (ANN), hyperparameter tuning is performed using the grid search (GS) algorithm. Grid search systematically explores a predefined set of hyperparameter values to find the combination that yields the best performance. The parameters optimized in this study include the following:

- **Number of Hidden Layers and Neurons:** The architecture of the ANN is fine-tuned to balance model complexity and computational efficiency.
- **Learning Rate:** The step size for weight updates is optimized to ensure efficient convergence.
- **Batch Size:** The number of training samples used in each iteration is adjusted to optimize training stability.
- **Activation Functions:** Various activation functions (e.g., ReLU, Sigmoid, Tanh) are evaluated for their impact on model performance.
- **Optimizer Selection:** In addition to hyperparameter tuning, the optimization algorithm itself is also optimized. Three different optimizers—Adam, RMSProp, and SGD—are evaluated to determine the most suitable choice for the model.

The optimized structure of the ANN is illustrated in Figure 4.

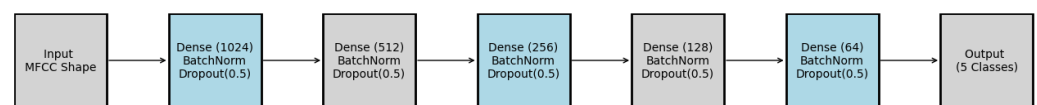


Figure 4. Structure-tuned ANN for the proposed method.

Phase 4: Model Training and Evaluation The final stage involves training and evaluating the ANN model using the Donate a Cry Dataset. The dataset is split into training and testing subsets, ensuring that the model is trained on a diverse set of crying sounds and evaluated on unseen samples.

The training process includes the following:

- **Weight Initialization:** The model weights are initialized using an appropriate strategy to prevent vanishing or exploding gradients.
- **Backpropagation and Optimization:** The loss function is minimized using the selected optimizer, and model weights are updated accordingly.
- **Performance Evaluation:** The trained model is tested on unseen crying sounds, and its classification accuracy, precision, recall, and F1-score are computed to assess its effectiveness.

Through these systematic steps, the proposed method ensures robust classification of baby crying sounds by leveraging data augmentation, feature extraction, hyperparameter tuning, and optimized model training.

3. Experimental Results

This paper used VGG16 (a pre-trained CNN model consisting of 16 layers, which is frequently used in image classification tasks and is an important milestone in the field of deep learning), KNN, CNN, random forest-based methods (M1, M2, M3, M4, M5, and M6), and the ANN-based proposed method to classify baby sounds. These methods were applied to analyze the sound data to classify baby sounds. These methods were applied to analyze audio data to classify baby sounds. It is aimed to use these methods by repeating studies in the literature with high accuracy rates. In the testing phase of the models, the Donate a Cry dataset was tested and compared to classify baby sounds. In the simulations,

the dataset was divided into three parts, 75% for training, 10% for validation, and 15% for testing, see the splitting in [24] for a comparison.

In the proposed method, hyperparameter optimization was performed using the grid search (GS) algorithm to tune the parameters of the ANN. The near-optimal hyperparameters selected by GS are shown in Table 5.

While using the grid search optimization algorithm, not only the hyperparameter optimization but also the optimization algorithm to be used is optimized. In this process, three different optimizer options are presented to the grid search algorithm: Adam (DeepMind, London, UK), RMSProp (Geoffrey Hinton, Toronto, ON, Canada), and SGD (Stochastic Gradient Descent, USA). Thus, the performance of the model is increased by determining the most appropriate optimizer and hyperparameter combination.

Table 5. Hyperparameter values of the ANN selected by GS.

Parameters	Value
Batch Size	32
Epoch	100
Optimizer	Adam
Activation	Relu
Dense Layers	5
Dropout	0.5

In Table 6, the results compare the performance of different models to classify baby voices. The models were supported by methods such as DA (Data Augmentation), MFCC (Mel-Frequency Cepstral Coefficients), and SF (Spectral Flatness) as sound features. Precision, recall, and F1 score values were examined as performance measures.

Table 6. Performance comparison of proposed method vs. the other generated models.

Method	Techniques	Metric	Belly Pain	Burping	Discomfort	Hungry	Tired
M1	RF + DA + MFCC	Precision	0.00%	0.00%	18.00%	83.00%	0.00%
		Recall	0.00%	0.00%	4.00%	88.00%	0.00%
		F1Score	0.00%	0.00%	8.00%	85.00%	0.00%
M2	One-vs-All + DA + SF	Precision	0.00%	0.00%	0.00%	83.00%	0.00%
		Recall	0.00%	0.00%	0.00%	100.00%	0.00%
		F1Score	0.00%	0.00%	0.00%	91.00%	0.00%
M3	KNN + DA + SF	Precision	0.00%	0.00%	50.00%	85.00%	0.00%
		Recall	0.00%	0.00%	4.00%	99.00%	0.00%
		F1Score	0.00%	0.00%	8.00%	91.00%	0.00%
M4	CNN + MFCC	Precision	0.00%	0.00%	0.00%	77.00%	0.00%
		Recall	0.00%	0.00%	0.00%	100.00%	0.00%
		F1Score	0.00%	0.00%	0.00%	87.00%	0.00%
M5	VGG16 + MFCC	Precision	0.00%	0.00%	0.00%	87.00%	0.00%
		Recall	0.00%	0.00%	0.00%	97.00%	0.00%
		F1Score	0.00%	0.00%	0.00%	92.00%	0.00%
M6	Alexnet + Split + MFCC	Precision	0.00%	0.00%	0.00%	87.00%	0.00%
		Recall	0.00%	0.00%	0.00%	100.00%	0.00%
		F1Score	0.00%	0.00%	0.00%	93.00%	0.00%
Proposed	ANN + GS + DA + MFCC	Precision	94.00%	100.00%	96.00%	90.00%	79.00%
		Recall	97.00%	100.00%	91.00%	81.00%	92.00%
		F1Score	95.00%	100.00%	93.00%	85.00%	85.00%

The poor performance of other methods on minority classes is likely class imbalance in the dataset. Traditional models such as random forest (RF), k-nearest neighbors (KNNs), and One-vs.-All tend to favor majority classes, as they rely on handcrafted features that might not sufficiently capture the subtle differences between cry categories. When the dataset is imbalanced, these models often learn to optimize for the dominant class, leading to a significant drop in performance for underrepresented classes.

Furthermore, traditional machine learning methods struggle with the complex and high-dimensional nature of baby cry sounds, which exhibit intricate variations in frequency, amplitude, and temporal patterns. These models typically depend on predefined statistical features, such as MFCCs or spectral features, which may not be expressive enough to distinguish minority classes effectively.

On the other hand, deep learning models, particularly artificial neural networks (ANNs), have the advantage of learning hierarchical and abstract feature representations directly from the data. By leveraging advanced optimization techniques and neural architectures, these models are able to capture nuanced differences between cry types, even in cases where certain classes have fewer training samples.

Additionally, our proposed method incorporates grid search (GS) for hyperparameter tuning, data augmentation (DA) to mitigate class imbalance, and MFCC-based feature extraction, all of which contribute to its superior performance. The ANN's ability to generalize well across different cry types, despite data imbalance, further highlights its effectiveness in this classification task. Future improvements could involve employing more sophisticated augmentation techniques or adaptive loss functions to further enhance performance across all classes.

When the results are examined, the highest accuracy (90%) and F1 score (90%) values were achieved with the structure-tuned artificial neural network (ANN)-based proposed method. The proposed method showed the best performance in terms of accuracy and F1 score, exhibiting a higher classification success than the other methods. Therefore, the proposed method model was preferred as the most appropriate solution for classifying baby sounds.

The 90% accuracy rate obtained was compared with the study that gave the best-known result in the literature. According to the results in Table 7, the proposed method left behind the most successful result in the same data splitting type.

Table 7. Comparison studies on Donate a Cry and accuracy rates.

Authors	Techniques	F1 Score	Accuracy
Jamal et al. [24]	ANN	46.99%	85.93%
Ours	ANN + GS + DA + MFCC	90%	90%

The similarities and differences between the proposed method for classifying baby sounds and the existing studies using the same feature extraction process are shown in Table 8.

In this study, the performance of the proposed method was evaluated in detail using a confusion matrix, learning curves, and ROC curves. The confusion matrix for the test data results is given in Figure 5. The graph measuring the best performance of the learning model over the epoch or iteration is shown in Figures 6 and 7.

The confusion matrix was used to evaluate the class-based accuracy and error rates of the model based on the test data. The majority of classes were correctly classified. The correct classification rates were high, especially for belly pain, burping, and discomfort (31, 16, and 49 correct classifications). Significant confusion was observed between hungry and tired. Hungry was frequently misclassified as tired (5 examples), and similarly, tired was

predicted as hungry in 7 examples. This situation shows that the features of these classes can be quite similar.

The learning curves showing the changes in the training and validation accuracies of the model across epochs are presented in Figure 6. The model learned rapidly on the training data for training accuracy and reached an accuracy rate of approximately 90%. However, this situation shows that the model is vulnerable to overfitting. For validation accuracy, although the accuracy on the validation dataset was initially low (30%), it increased over time and reached approximately 60%. However, the significant difference between the training and validation accuracy reveals that the model is poor at generalizing the validation data.

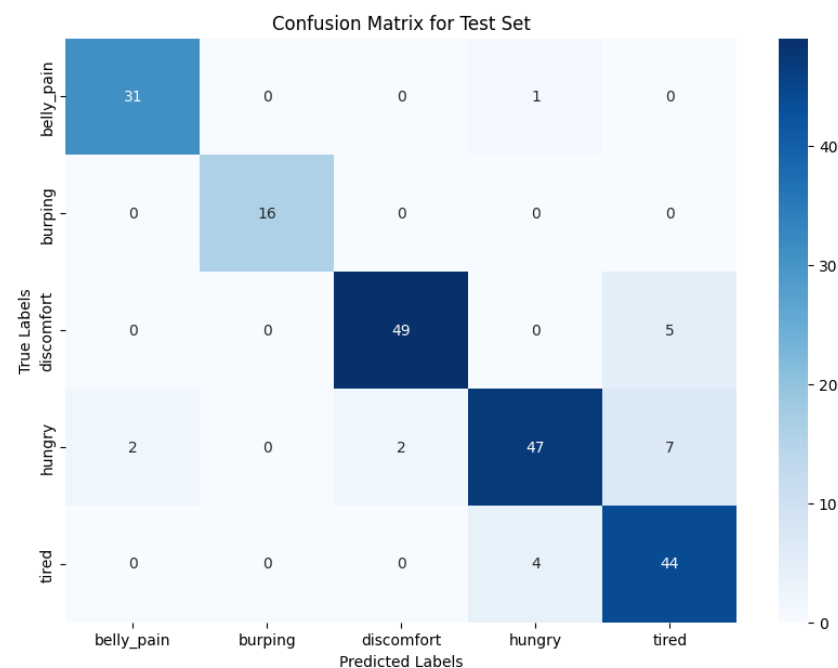


Figure 5. Confusion matrix for the proposed method.

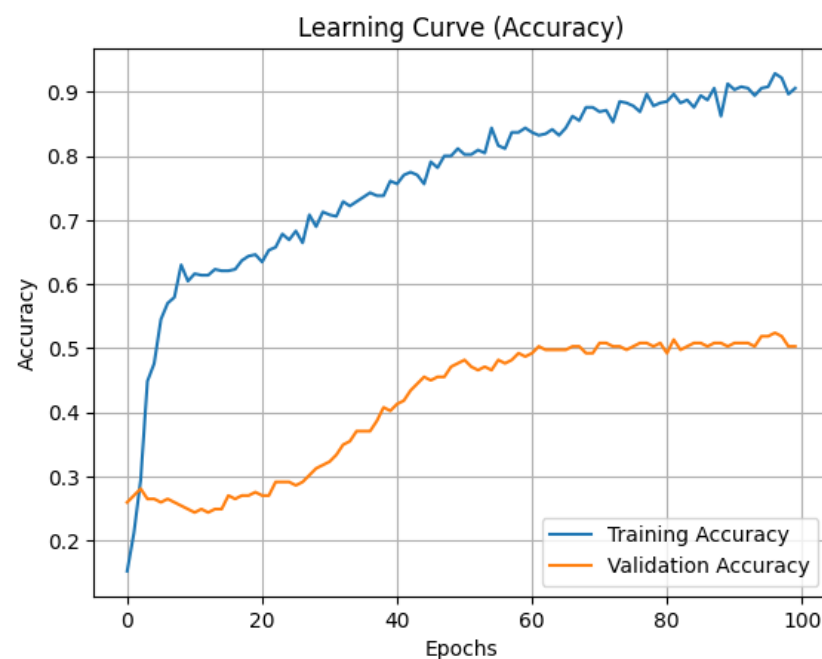


Figure 6. Learning curves for ANN.

Table 8. Comparison of studies using MFCC feature extraction with the proposed method.

First author	Dataset	Features	Classifiers	Differences	Similarity
Liu [6]	NICU recorded (draw attention, cry, diaper change needed, cry, and hungry)	LPC, LPCC, MFCC, BFCC	KNN, ANN	Different dataset and classifiers	Baby cry classification Same feature extraction method
Bano [10]	Self-recorded (hungry, need to burp, sleepy, pain, discomfort)	Pitch short-time energy MFCC Statistical properties of MFCC	KNN	Different dataset and classifiers	Baby cry classification Same feature extraction method
Bhagatpatil [12]	Self-recorded (pain, hunger, discomfort, need to burp, belly pain)	LFCC, MFCC	KNN	Different dataset and classifiers	Baby cry classification Same feature extraction method
Rosales-Pérez [13]	Baby Chillanto (hungry vs. pain)	MFCC, LPC	Fuzzy model	Different dataset and classifiers	Baby cry classification Same Feature extraction method
Bashiri [15]	Baby Chillanto	MFCC	ANN	Different dataset	Baby cry classification Same feature extraction method and classifiers
Manikanta [17]	Recorded in homes (crying with ac, fan, etc.)	MFCC	1D-CNN, FFNN, SVM	Baby cry detection Different dataset and classifiers	Same feature extraction method
Ferretti [20]	Real Dataset: recorded in the NICU of a hospital; Synthetic DB: crying with speech, “beep” sounds, etc.)	Log-Mel Coefficients	CNN	Baby cry detection Different dataset and classifiers	Same feature extraction method
Lavner [23]	Recorded in domestic environment (crying with speech, door opening, etc.)	MFCC, Pitch, Formants, etc.	CNN	Baby cry detection Different dataset and classifiers	Same Feature extraction method
Jamal et al. [24]	Baby Cry and Donate a Cry	Chroma Deviation	ANN	Different feature extraction method	Baby cry classification Same dataset and classifiers
Proposed Method	Donate a Cry	MFCC	ANN	Data augmentation techniques Hyperparameters optimization	Baby cry classification Same dataset, classifiers and feature extraction method

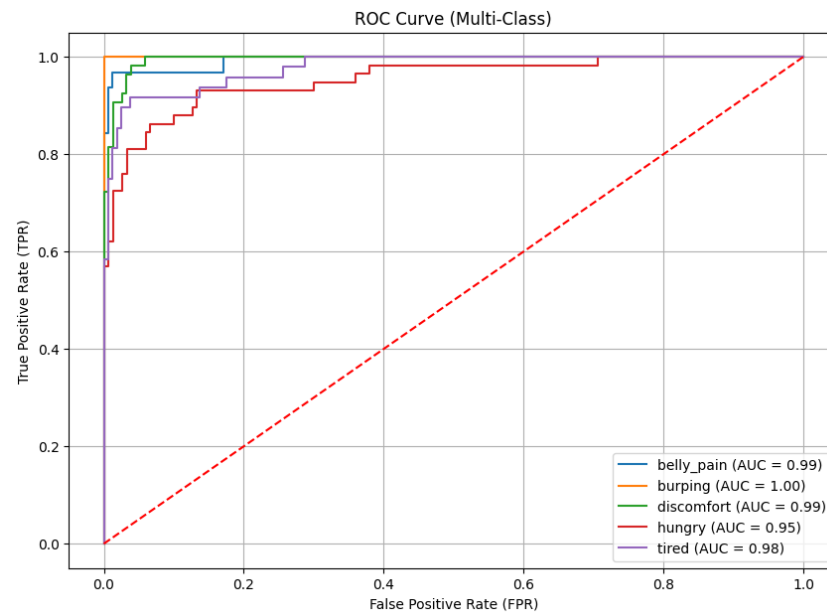


Figure 7. ROC curve for ANN.

The ROC curve presented in Figure 7 was used to evaluate the model's inter-class discrimination power. The AUC values for all classes are quite high (0.99 for Belly pain, 1.00 for Burping, 0.99 for Discomfort, 0.95 for Hungry, and 0.98 for Tired), which shows that the model has good discrimination power in general. The One-vs-Rest (OvR) approach was used in calculating the ROC curves.

The slight decreases in AUC values, especially for hungry and tired (0.95 and 0.98, respectively), indicate a result consistent with the confusion between these classes. Feature engineering and/or dataset balancing are recommended to separate these classes better.

To analyze the robustness of the proposed method in this article, the 10-fold cross-validation method was also applied. The dataset was divided into 10 different subsets, and the model was trained and tested in each of them. After the model training was completed in each layer, predictions were made on the validation set, and the predicted class labels were stored. Finally, the model's overall performance was calculated by combining the results of all layers.

The results show the model's performance against different classes and measure the classification quality. The model accuracy is presented with the average accuracy calculated as a result of cross-validation. As a result of 10-fold cross-validation, an accuracy rate of 87% was obtained.

4. Conclusions

In this paper, different artificial intelligence models were applied and tested to classify baby sounds. The performances of pre-trained VGG16, KNN, CNN, random forest, and artificial neural network (ANN) models were evaluated using the Donate a Cry dataset. The success rates of the models increased thanks to the hyperparameter optimizations made with the grid search algorithm. The results showed that the test accuracy reached 90% in the proposed method, especially obtained using a five-layer artificial neural network (ANN). This success rate reveals the effectiveness of deep learning methods in correctly classifying baby crying sounds.

The important outcomes of the study include the effectiveness of artificial intelligence models, data processing techniques, and augmentation methods. The proposed method provided high accuracy in classifying baby sounds. In addition, features such as Mel-Frequency Cepstral Coefficients (MFCCs), time, and spectral domain features significantly

increased the classification performance. Data augmentation techniques, such as time shifting, pitch shifting, and white noise augmentation methods, enriched the dataset and strengthened the model's generalization ability.

The results of this study have potential applications in health services and infant care. In particular, they can contribute to the development of real-time systems that can analyze baby crying sounds to provide fast and accurate information to parents or healthcare professionals.

Future work can focus on testing models using larger datasets and developing systems optimized for real-time applications. Furthermore, using different acoustic features and more advanced deep learning architectures can further improve classification performance.

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Abbreviations

The following abbreviations are used in this manuscript:

ANN	Artificial Neural Network
AUC	Area Under Curve
BFCCs	Bark-Frequency Cepstral Coefficients
CNN	Convolutional Neural Network
CRIED	Cry Recognition and Infant Emotion Detection dataset
CRNN	Convolutional Recurrent Neural Network
cu112	CUDA Toolkit Version 11.2
DA	Data Augmentation
DCT	Discrete Cosine Transform
F0	Fundamental Frequency
F123	First, Second, and Third Formant Frequencies
FFT	Fast Fourier Transform
GS	Grid Search
KNN	k-Nearest Neighbors
LFCC	Linear Frequency Cepstral Coefficient
LPC	Linear Predictive Coding
LPCC	Linear Predictive Cepstral Coefficient
LogSoftmax	Logarithmic Softmax Activation
M1	Random Forest, Data Augmentation, Mel-Frequency Cepstral Coefficients
M2	One-versus-All, Data Augmentation, Spectral Flatness
M3	k-Nearest Neighbors, Data Augmentation, Spectral Flatness
M4	Convolutional Neural Network, Mel-Frequency Cepstral Coefficients
M5	Visual Geometry Group 16-layer network, Mel-Frequency Cepstral Coefficients
M6	Alexnet, Data Split, Mel-Frequency Cepstral Coefficient
MFCC	Mel-Frequency Cepstral Coefficient
NICU	Neonatal Intensive Care Unit

ODbL	Open Database License
OvR	One-vs.-Rest
ReLU	Rectified Linear Unit
RF	Random Forest
RMS	Root Mean Square
ROC	Receiver Operating Characteristic
RNN	Recurrent Neural Network
SF	Spectral Flatness
STFT	Short-Time Fourier Transform
SVDD	Support Vector Data Descriptor
SVM	Support Vector Machine
VGG16	Visual Geometry Group 16-layer network
ZCR	Zero Crossing Rate

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